Министерство образования Республики Беларусь
Учреждение образования
«Брестский государственный технический университет»
Кафедра ИИТ

**Лабораторная работа №3**
По дисциплине: «ОИвИС»
Тема: «Обучение детекторов объектов»

**Выполнил:**
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Брест 2026

**Цель работы:**

осуществлять обучение нейросетевого детектора для решения задачи обнаружения заданных объектов

**Общее задание**

1. Базируясь на своем варианте, ознакомится с выборкой для обучения детектора, выполнить необходимые преобразования данных для организации процесса обучения (если это нужно!);
2. Для заданной архитектуры нейросетевого детектора организовать процесс обучения для своей выборки. Оценить эффективность обучения на тестовой выборке (mAP);
3. Реализовать визуализацию работы детектора из пункта 1 (обнаружение знаков на отдельных фотографиях из сети Интернет);
4. Оформить отчет по выполненной работе, залить исходный код и отчет в соответствующий репозиторий на github.

**Вариант 8**

| | | |
|---|---|---|
| 8 | YOLO12n | **Номерные знаки авто:** https://universe.roboflow.com/roboflow-universe-projects/license-plate-recognition-rxg4e/dataset/11 |

**Код программы:**

```
import random
from pathlib import Path
import yaml
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.patches as patches
from PIL import Image
import torch
import cv2
from ultralytics import YOLO
from roboflow import Roboflow

random.seed(42)


class PlateDetectorPipeline:
    def __init__(self, api_key="QUYMcDyS147x6eWX6IUw"):
        self.api_key = api_key
        self.rf = Roboflow(api_key=self.api_key)
        self.proj = self.rf.workspace("roboflow-universe-projects").project("license-plate-
recognition-rxg4e")
        self.ds = self.proj.version(11).download("yolov8")
        self.root = Path(self.ds.location)
        self.yaml_path = self.root / "data.yaml"
        self._prepare_yaml()
        self._report_counts()
        self.model = None
        self.save_dir = None
        self.best_model = None
```

```
def _prepare_yaml(self):
    with open(self.yaml_path) as f:
        cfg = yaml.safe_load(f)
    print(f"Classes ({len(cfg['names'])}): {cfg['names']}")
    cfg["path"] = str(self.root)
    cfg["train"] = "train/images"

    valid_imgs = self.root / "valid" / "images"
    if valid_imgs.exists() and any(valid_imgs.glob("*")):
        cfg["val"] = "valid/images"
    else:
        print("valid/ пустая — использую test/images как val")
        cfg["val"] = "test/images"

    cfg["test"] = "test/images"

    with open(self.yaml_path, "w") as f:
        yaml.safe_dump(cfg, f, allow_unicode=True)
    self.cfg = cfg

def _report_counts(self):
    for part in ("train", "valid", "test"):
        p = self.root / part / "images"
        cnt = sum(1 for _ in p.glob("*")) if p.exists() else 0
        print(f"{part}: {cnt}")

def preview_batch(self, part="train", k=6):
    files = random.sample(list((self.root / part / "images").glob("*")), k)
    fig, axs = plt.subplots(2, 3, figsize=(14, 8))
    for ax, fp in zip(axs.flat, files):
        im = Image.open(fp).convert("RGB")
        w, h = im.size
        ax.imshow(im)
        ax.axis("off")
        lbl = self.root / part / "labels" / (fp.stem + ".txt")
        if lbl.exists():
            for ln in lbl.read_text().strip().splitlines():
                parts = ln.split()
                if len(parts) < 5:
                    continue
                _, xc, yc, bw, bh = map(float, parts[:5])
                ax.add_patch(patches.Rectangle(
                    ((xc - bw / 2) * w, (yc - bh / 2) * h),
                    bw * w, bh * h, fill=False, edgecolor="lime", linewidth=2))
    plt.suptitle("Sample training images with labels")
    plt.tight_layout()
    plt.show()

def train(self, epochs=20):
    self.model = YOLO("yolo12n.pt")
    self.model.train(
        data=str(self.yaml_path),
        epochs=epochs,
        imgsz=640,
        batch=16,
        project="plate_training",
        name="run1",
        exist_ok=True,
        seed=42,
    )
    self.save_dir = Path(self.model.trainer.save_dir)
    print(f"Training output: {self.save_dir}")

def plot_history(self):
    log = pd.read_csv(self.save_dir / "results.csv")
```

```
        log.columns = log.columns.str.strip()
        fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(13, 5))

        ax1.plot(log["epoch"], log["train/box_loss"], label="train", marker="s", linestyle="--
", markersize=3)
        ax1.plot(log["epoch"], log["val/box_loss"], label="val", marker="s", linestyle="--",
markersize=3)
        ax1.set_xlabel("epoch")
        ax1.set_ylabel("loss")
        ax1.set_title("Box loss")
        ax1.legend()
        ax1.grid(alpha=0.3)

        ax2.plot(log["epoch"], log["metrics/mAP50(B)"], label="mAP@0.5", marker="^",
linestyle="-.", markersize=3)
        ax2.plot(log["epoch"], log["metrics/mAP50-95(B)"], label="mAP@0.5:0.95", marker="^",
linestyle="-.",
                 markersize=3)
        ax2.set_xlabel("epoch")
        ax2.set_ylabel("mAP")
        ax2.set_title("Validation mAP")
        ax2.legend()
        ax2.grid(alpha=0.3)

        plt.tight_layout()
        plt.show()

    def evaluate(self):
        self.best_model = YOLO(str(self.save_dir / "weights" / "best.pt"))
        m = self.best_model.val(data=str(self.yaml_path), split="test", imgsz=640)
        print(f"P={m.box.mp:.4f}, R={m.box.mr:.4f}, mAP50={m.box.map50:.4f}, mAP50-
95={m.box.map:.4f}")
        return m

    def process_videos(self, sources, dst="output_combined.mp4", conf=0.2, show=False):
        caps = [cv2.VideoCapture(str(s)) for s in sources]
        for i, cap in enumerate(caps):
            if not cap.isOpened():
                raise FileNotFoundError(f"Не открывается видео: {sources[i]}")

        fps = caps[0].get(cv2.CAP_PROP_FPS) or 25
        w = int(caps[0].get(cv2.CAP_PROP_FRAME_WIDTH))
        h = int(caps[0].get(cv2.CAP_PROP_FRAME_HEIGHT))

        writer = cv2.VideoWriter(str(dst), cv2.VideoWriter_fourcc(*"mp4v"), fps, (w, h))

        total_frames = 0
        total_dets = 0

        for cap, src_name in zip(caps, sources):
            print(f"--- Обработка: {src_name} ---")
            while True:
                ok, frame = cap.read()
                if not ok:
                    break

                if frame.shape[1] != w or frame.shape[0] != h:
                    frame = cv2.resize(frame, (w, h))

                res = self.best_model.predict(frame, conf=conf, imgsz=640, max_det=100,
verbose=False)[0]

                for box in res.boxes:
                    x1, y1, x2, y2 = map(int, box.xyxy[0].tolist())
                    score = float(box.conf[0])
```

```
                    cv2.rectangle(frame, (x1, y1), (x2, y2), (0, 255, 0), 2)
                    cv2.putText(frame, f"plate {score:.2f}", (x1, max(y1 - 5, 15)),
cv2.FONT_HERSHEY_SIMPLEX, 0.6,
                                (0, 255, 0), 2)
                    total_dets += 1

                writer.write(frame)
                total_frames += 1

                if show:
                    cv2.imshow("detect", frame)
                    if cv2.waitKey(1) & 0xFF == 27:
                        break

            cap.release()

        writer.release()
        if show:
            cv2.destroyAllWindows()

        print(f"Готово! Кадров обработано: {total_frames}, детекций: {total_dets}")
        print(f"Единое видео сохранено: {Path(dst).resolve()}")

    def run_all(self):
        self.preview_batch("train")
        self.train(20)
        self.plot_history()
        self.evaluate()

        video_paths = [
            r"C:\Users\arosl\PycharmProjects\OIS3\1.mp4",
            r"C:\Users\arosl\PycharmProjects\OIS3\2.mp4"
        ]

        self.process_videos(video_paths, "output_combined.mp4")

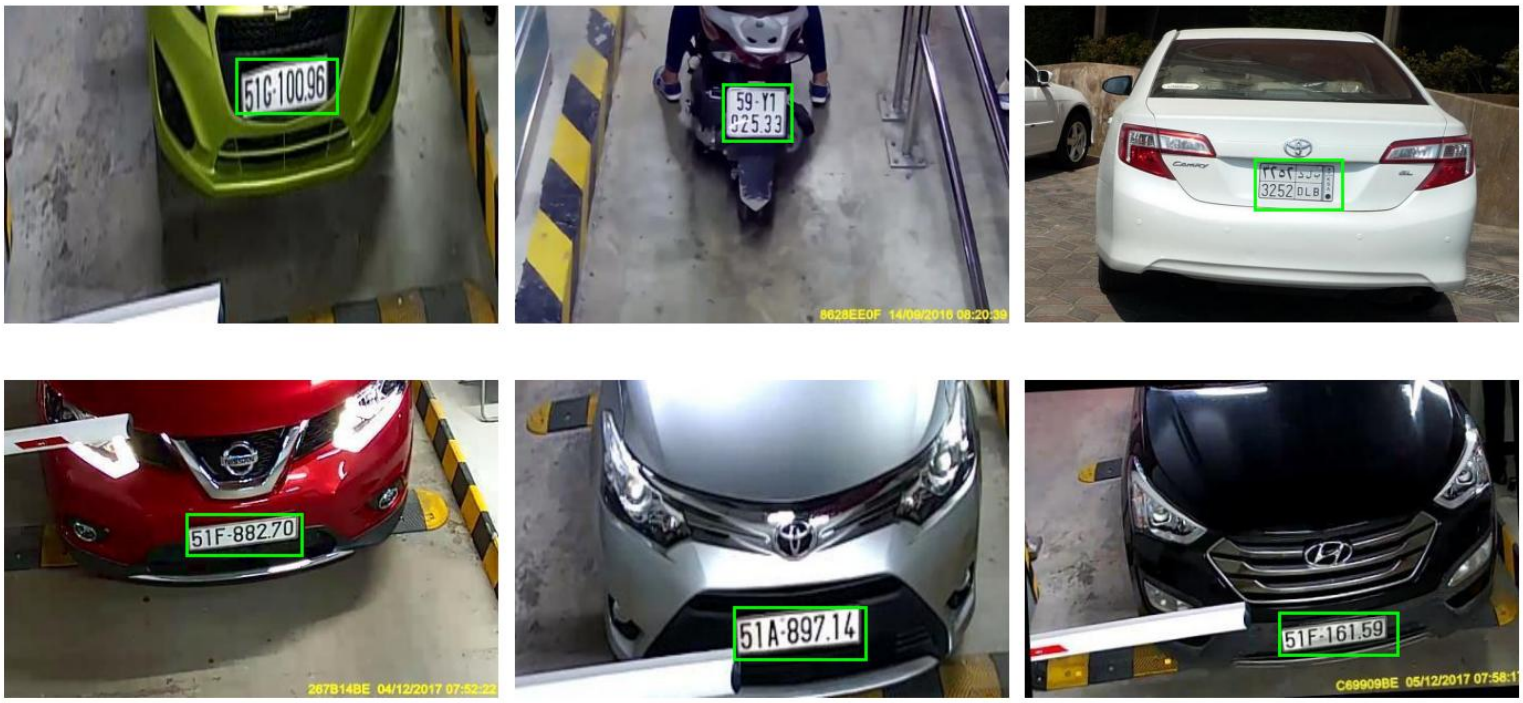

if __name__ == "__main__":
    pipe = PlateDetectorPipeline()
    pipe.run_all()
```

## Результат:

```
Creating new Ultralytics Settings v0.0.8 file ☑
View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For
help see https://docs.ultralytics.com/usage/settings.
loading Roboflow workspace...
loading Roboflow project...
Downloading Dataset Version Zip in License-Plate-Recognition-11 to yolov8:: 100%|██████████|
527254/527254 [00:10<00:00, 49930.89it/s]


Extracting Dataset Version Zip to License-Plate-Recognition-11 in yolov8:: 100%|██████████|
20262/20262 [00:02<00:00, 7323.73it/s]
Classes (1): ['License_Plate']
train: 7057
valid: 2048
test: 1020
```

Sample training images with labels




Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo12n.pt to 'yolo12n.pt': 100% ━━━━━━━━━━━━━━━━━━━━ 5.3MB 97.4MB/s 0.1s

Ultralytics 8.4.178 🚀 Python-3.13.15 torch-2.11.0+cu130 CUDA:0 (Tesla T4, 14913MiB)

**engine/trainer:** agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augment=randaugment, batch=16, bgr=0.0, box=7.5, cache=False, cfg=None, channels_last=None, classes=None, close_mosaic=10, cls=0.5, cls_pw=0.0, cls_remap=True, compile=False, conf=None, copy_paste=0.0, copy_paste_mode=flip, cos_lr=False, cutmix=0.0, data=/content/License-Plate-Recognition-11/data.yaml, degrees=0.0, deterministic=True, device=, dfl=1.5, dgrad=0.5, dis=6.0, distill_model=None, dlam=1.0, dlog=1.0, dnn=False, dropout=0.0, dynamic=False, embed=None, epochs=20, erasing=0.4, exist_ok=True, fliplr=0.5, flipud=0.0, format=torchscript, fraction=1.0, freeze=None, hsv_h=0.015, hsv_s=0.7, hsv_v=0.4, imgsz=640, iou=0.7, kobj=1.0, line_width=None, lr0=0.01, lrf=0.01, mask_ratio=4, max_det=300, mixup=0.0, mode=train, model=yolo12n.pt, momentum=0.937, mosaic=1.0, multi_scale=0.0, name=run1, nbs=64, nms=None, opset=None, optimize=False, optimizer=auto, overlap_mask=True, patience=100, perspective=0.0, plots=True, pose=12.0, pretrained=True, profile=False, project=plate_training, quantize=None, rect=False, resume=False, retina_masks=False, rle=1.0, save=True, save_conf=False, save_crop=False, save_dir=/content/runs/detect/plate_training/run1, save_frames=False, save_json=False, save_period=-1, save_txt=False, scale=0.5, seed=42, shear=0.0, show=False, show_boxes=True, show_conf=True, show_labels=True, simplify=True, single_cls=False, source=None, split=val, stream_buffer=False, task=detect, time=None, tracker=tracktrack.yaml, translate=0.1, val=True, verbose=True, vid_stride=1, visualize=False, warmup_bias_lr=0.1, warmup_epochs=3.0, warmup_momentum=0.8, weight_decay=0.0005, workers=8, workspace=None

Downloading https://ultralytics.com/assets/Arial.ttf to '/root/.config/Ultralytics/Arial.ttf': 100% ━━━━━━━━━━━━━━━━━━━━ 755.1KB 25.4MB/s 0.0s

Overriding model.yaml nc=80 with nc=1

| | from | n | params | module | arguments |
|---|---|---|---|---|---|
| 0 | -1 | 1 | 464 | ultralytics.nn.modules.conv.Conv | [3, 16, 3, 2] |
| 1 | -1 | 1 | 4672 | ultralytics.nn.modules.conv.Conv | [16, 32, 3, 2] |
| 2 | -1 | 1 | 6640 | ultralytics.nn.modules.block.C3k2 | [32, 64, 1, False, 0.25] |
| 3 | -1 | 1 | 36992 | ultralytics.nn.modules.conv.Conv | [64, 64, 3, 2] |
| 4 | -1 | 1 | 26080 | ultralytics.nn.modules.block.C3k2 | [64, 128, 1, False, 0.25] |

```
  5                  -1  1    147712  ultralytics.nn.modules.conv.Conv             [128, 128,
3, 2]
  6                  -1  2    180864  ultralytics.nn.modules.block.A2C2f           [128, 128,
2, True, 4]
  7                  -1  1    295424  ultralytics.nn.modules.conv.Conv             [128, 256,
3, 2]
  8                  -1  2    689408  ultralytics.nn.modules.block.A2C2f           [256, 256,
2, True, 1]
  9                  -1  1         0  torch.nn.modules.upsampling.Upsample         [None, 2,
'nearest']
 10             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 11                 -1  1     86912  ultralytics.nn.modules.block.A2C2f           [384, 128,
1, False, -1]
 12                 -1  1         0  torch.nn.modules.upsampling.Upsample         [None, 2,
'nearest']
 13             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 14                 -1  1     24000  ultralytics.nn.modules.block.A2C2f           [256, 64, 1,
False, -1]
 15                 -1  1     36992  ultralytics.nn.modules.conv.Conv             [64, 64, 3,
2]
 16            [-1, 11]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 17                 -1  1     74624  ultralytics.nn.modules.block.A2C2f           [192, 128,
1, False, -1]
 18                 -1  1    147712  ultralytics.nn.modules.conv.Conv             [128, 128,
3, 2]
 19             [-1, 8]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 20                 -1  1    378880  ultralytics.nn.modules.block.C3k2            [384, 256,
1, True]
 21        [14, 17, 20]  1    430867  ultralytics.nn.modules.head.Detect           [1, 16,
None, [64, 128, 256]]
YOLOv12n summary: 272 layers, 2,568,243 parameters, 2,568,227 gradients, 7.5 GFLOPs

Transferred 685/691 items from pretrained weights
Freezing layer 'model.21.dfl.conv.weight'
AMP: running Automatic Mixed Precision (AMP) checks...
AMP: downloading yolo26n.pt for AMP checks (one-time, not used for training)...
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo26n.pt to
'weights/yolo26n.pt': 100% ━━━━━━━━━━━━━━━━━━━━ 5.3MB 87.6MB/s 0.1s
AMP: checks passed ☑
train: Fast image access ☑ (ping: 0.0±0.0 ms, read: 836.8±364.7 MB/s, size: 32.3 KB)
train: Scanning /content/License-Plate-Recognition-11/train/labels... 7057 images, 5
backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━━━━━━━━━ 7057/7057 1.9Kit/s 3.6s
train: New cache created: /content/License-Plate-Recognition-11/train/labels.cache
albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)),
ToGray(p=0.01, method='weighted_average', num_output_channels=3), CLAHE(p=0.01,
clip_limit=(1.0, 4.0), tile_grid_size=(8, 8))
val: Fast image access ☑ (ping: 0.0±0.0 ms, read: 538.3±212.2 MB/s, size: 17.2 KB)
val: Scanning /content/License-Plate-Recognition-11/valid/labels... 2048 images, 3 backgrounds,
0 corrupt: 100% ━━━━━━━━━━━━━━━━━━━━ 2048/2048 2.1Kit/s 1.0s
val: New cache created: /content/License-Plate-Recognition-11/valid/labels.cache
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and determining best 'optimizer' and
'lr0' automatically...
optimizer: AdamW(lr=0.002, momentum=0.9) with parameter groups 113 weight(decay=0.0), 120
weight(decay=0.0005), 119 bias(decay=0.0)
Plotting labels to /content/runs/detect/plate_training/run1/labels.jpg...
Using 7057 train, 2048 val images for fraction=1.0 at imgsz=640
Using 2 dataloader workers
Logging results to /content/runs/detect/plate_training/run1
Starting training for 20 epochs...

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       1/20      3.38G      1.313      1.223      1.187          1        640: 100% ━━━━━━━━━━
━━━━━━━━ 442/442 3.6it/s 2:02
```

```
               Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 64/64 3.9it/s 16.3s
                 all       2048       2195      0.915      0.843      0.897      0.571

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       2/20      4.62G      1.315     0.8419      1.198          3        640: 100% ━━━━━━━━━━━━
━━━━━━━━ 442/442 4.0it/s 1:50
               Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 64/64 4.5it/s 14.1s
                 all       2048       2195      0.899      0.851      0.881      0.519

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       3/20      4.62G       1.29     0.8102      1.195          4        640: 100% ━━━━━━━━━━━━
━━━━━━━━ 442/442 4.0it/s 1:51
               Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 64/64 4.6it/s 13.9s
                 all       2048       2195      0.866      0.844      0.871      0.537

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       4/20      4.62G      1.272     0.7575      1.187          1        640: 100% ━━━━━━━━━━━━
━━━━━━━━ 442/442 3.9it/s 1:53
               Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 64/64 4.6it/s 14.0s
                 all       2048       2195      0.925       0.88      0.908      0.577

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       5/20      4.62G       1.26     0.7182      1.178          4        640: 100% ━━━━━━━━━━━━
━━━━━━━━ 442/442 3.9it/s 1:53
               Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 64/64 4.6it/s 13.9s
                 all       2048       2195      0.957      0.903      0.944      0.617

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       6/20      4.62G      1.224     0.6882      1.153          1        640: 100% ━━━━━━━━━━━━
━━━━━━━━ 442/442 3.9it/s 1:52
               Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 64/64 4.5it/s 14.1s
                 all       2048       2195      0.967      0.901      0.947      0.616

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       7/20      4.62G      1.199     0.6512      1.137          0        640: 100% ━━━━━━━━━━━━
━━━━━━━━ 442/442 4.0it/s 1:51
               Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 64/64 4.6it/s 13.9s
                 all       2048       2195      0.961      0.918      0.948      0.628

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       8/20      4.62G      1.183     0.6351      1.135          1        640: 100% ━━━━━━━━━━━━
━━━━━━━━ 442/442 3.9it/s 1:52
               Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 64/64 4.6it/s 13.8s
                 all       2048       2195      0.969      0.907      0.946      0.634

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
       9/20      4.62G      1.174     0.6143      1.134          2        640: 100% ━━━━━━━━━━━━
━━━━━━━━ 442/442 3.9it/s 1:53
               Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 64/64 4.8it/s 13.4s
                 all       2048       2195      0.977      0.909      0.952      0.643

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
```

```
      10/20      4.62G      1.172        0.6      1.123          1        640: 100% ━━━━━━━━━━━━
━━━━━━━ 442/442 3.9it/s 1:53
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 64/64 4.8it/s 13.4s
                   all       2048       2195      0.977      0.922      0.956      0.654
Closing dataloader mosaic
albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01, blur_limit=(3, 7)),
ToGray(p=0.01, method='weighted_average', num_output_channels=3), CLAHE(p=0.01,
clip_limit=(1.0, 4.0), tile_grid_size=(8, 8))

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      11/20      4.62G      1.142     0.5511      1.126          1        640: 100% ━━━━━━━━━━━━
━━━━━━━ 442/442 4.0it/s 1:49
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 64/64 4.8it/s 13.4s
                   all       2048       2195      0.974      0.917      0.954      0.643

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      12/20      4.62G      1.144     0.5336      1.132          1        640: 100% ━━━━━━━━━━━━
━━━━━━━ 442/442 4.2it/s 1:45
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 64/64 4.8it/s 13.3s
                   all       2048       2195      0.974      0.917      0.957       0.65

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      13/20      4.62G       1.12     0.5173      1.115          1        640: 100% ━━━━━━━━━━━━
━━━━━━━ 442/442 4.2it/s 1:45
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 64/64 4.7it/s 13.5s
                   all       2048       2195      0.975      0.923      0.962      0.657

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      14/20      4.62G      1.105     0.4948      1.109          1        640: 100% ━━━━━━━━━━━━
━━━━━━━ 442/442 4.2it/s 1:45
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 64/64 4.7it/s 13.5s
                   all       2048       2195      0.977      0.926       0.96      0.657

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      15/20      4.62G      1.099     0.4872      1.105          1        640: 100% ━━━━━━━━━━━━
━━━━━━━ 442/442 4.2it/s 1:46
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 64/64 4.8it/s 13.2s
                   all       2048       2195       0.98      0.934      0.968      0.665

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      16/20      4.62G      1.087     0.4714      1.097          1        640: 100% ━━━━━━━━━━━━
━━━━━━━ 442/442 4.2it/s 1:45
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 64/64 4.8it/s 13.5s
                   all       2048       2195      0.978      0.936      0.961       0.67

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      17/20      4.62G      1.072     0.4602      1.087          1        640: 100% ━━━━━━━━━━━━
━━━━━━━ 442/442 4.2it/s 1:44
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 64/64 4.8it/s 13.4s
                   all       2048       2195      0.981      0.922      0.959      0.667

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
```

```
      18/20      4.62G      1.058     0.4432      1.085          1        640: 100% ━━━━━━━━━━━━
━━━━━━━ 442/442 4.2it/s 1:45
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 64/64 4.8it/s 13.3s
                   all       2048       2195      0.983      0.939      0.968      0.672

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      19/20      4.62G       1.05     0.4292      1.075          1        640: 100% ━━━━━━━━━━━━
━━━━━━━ 442/442 4.2it/s 1:44
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 64/64 4.9it/s 13.0s
                   all       2048       2195      0.984      0.944       0.97      0.677

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
      20/20      4.62G      1.041     0.4205      1.067          1        640: 100% ━━━━━━━━━━━━
━━━━━━━ 442/442 4.2it/s 1:45
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 64/64 4.8it/s 13.3s
                   all       2048       2195      0.986      0.944      0.971       0.68

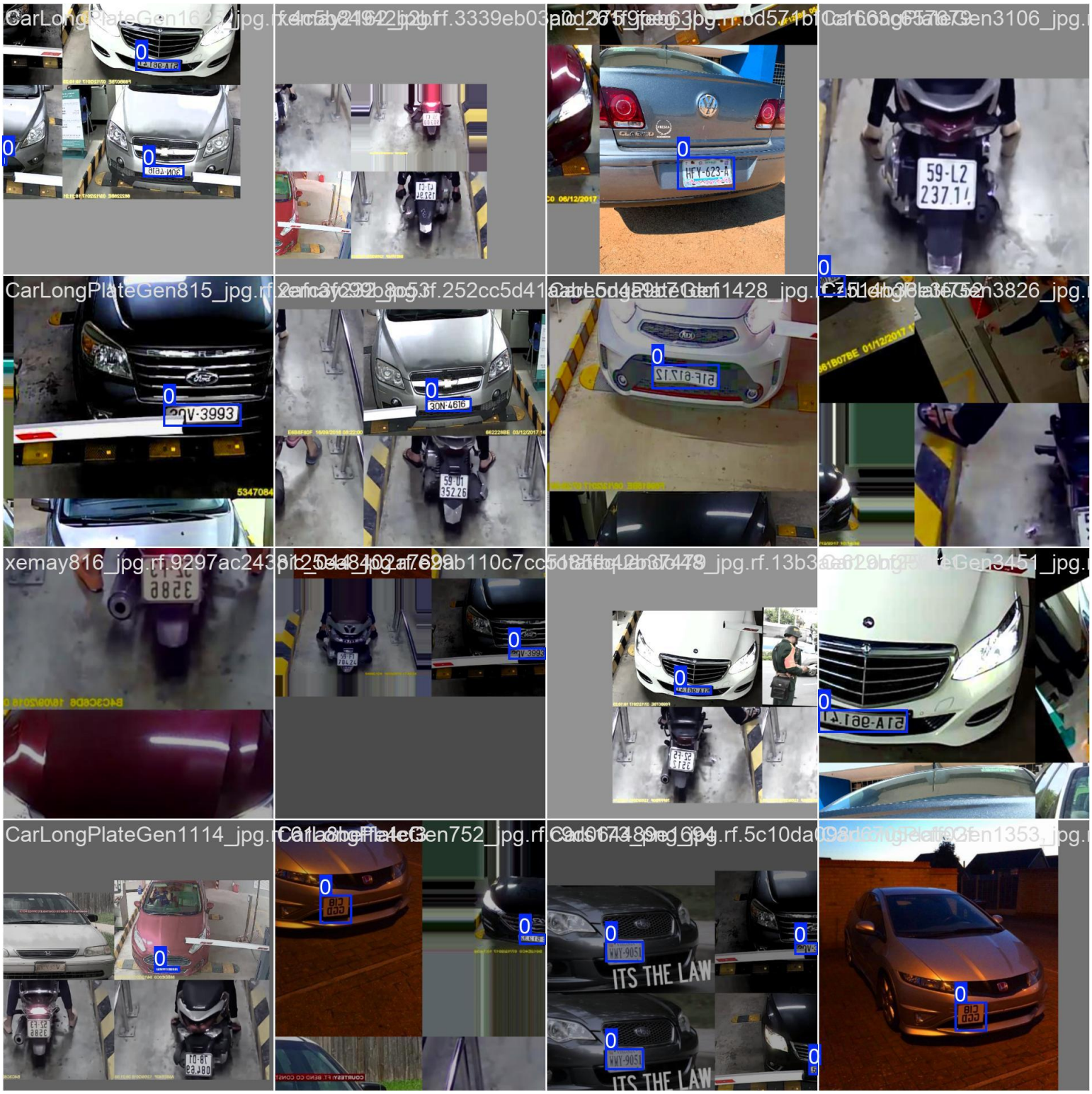
20 epochs completed in 0.687 hours.
Optimizer stripped from /content/runs/detect/plate_training/run1/weights/last.pt, 5.5MB
Optimizer stripped from /content/runs/detect/plate_training/run1/weights/best.pt, 5.5MB

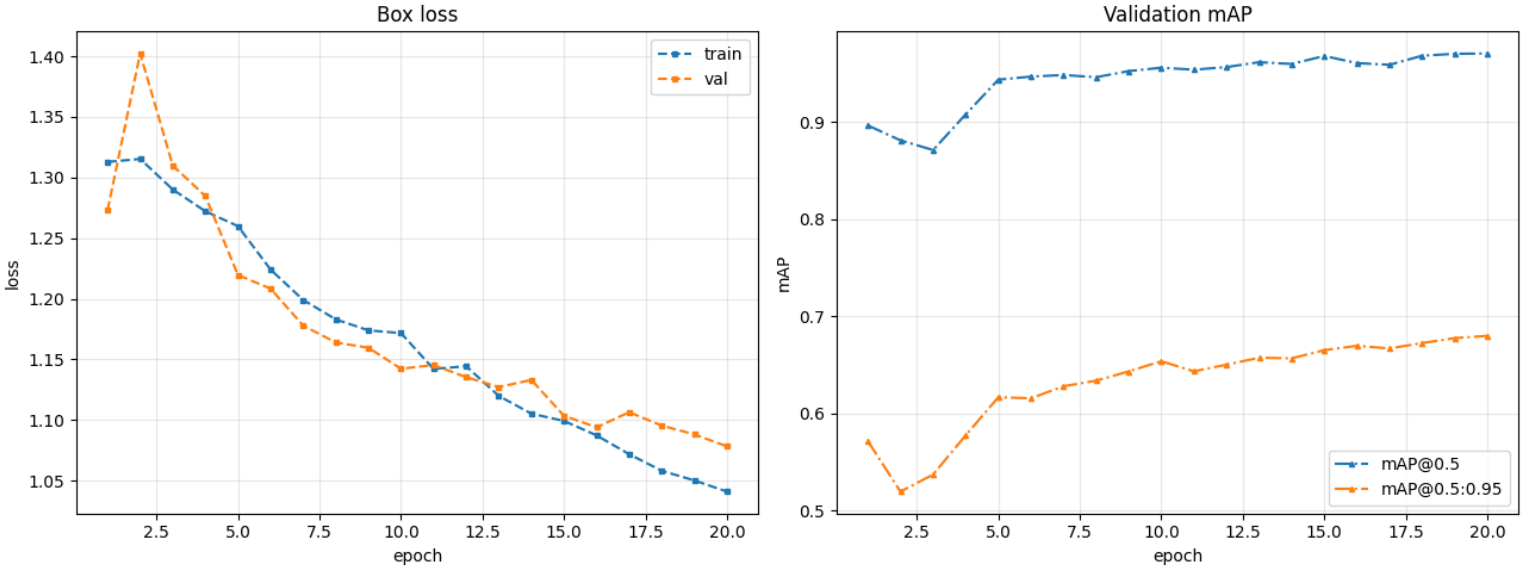
Validating /content/runs/detect/plate_training/run1/weights/best.pt...
Ultralytics 8.4.178 🚀 Python-3.13.15 torch-2.11.0+cu130 CUDA:0 (Tesla T4, 14913MiB)
YOLOv12n summary (fused): 160 layers, 2,571,627 parameters, 0 gradients, 7.4 GFLOPs
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 64/64 3.4it/s 19.0s
                   all       2048       2195      0.986      0.944      0.971       0.68
Speed: 0.2ms preprocess, 4.6ms inference, 0.0ms loss, 1.2ms postprocess per image
Results saved to /content/runs/detect/plate_training/run1
Training output: /content/runs/detect/plate_training/run1

Ultralytics 8.4.178 🚀 Python-3.13.15 torch-2.11.0+cu130 CUDA:0 (Tesla T4, 14913MiB)
YOLOv12n summary (fused): 160 layers, 2,571,627 parameters, 0 gradients, 7.4 GFLOPs
val: Fast image access ☑ (ping: 0.0±0.0 ms, read: 1122.8±495.4 MB/s, size: 69.7 KB)
val: Scanning /content/License-Plate-Recognition-11/test/labels... 1020 images, 1 backgrounds,
0 corrupt: 100% ━━━━━━━━━━━━━━━━━━━━ 1020/1020 2.1Kit/s 0.5s
val: New cache created: /content/License-Plate-Recognition-11/test/labels.cache
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 64/64 6.0it/s 10.7s
                   all       1020       1085       0.98      0.953       0.97       0.69
Speed: 1.2ms preprocess, 4.7ms inference, 0.0ms loss, 1.4ms postprocess per image
Results saved to /content/runs/detect/val
P=0.9799, R=0.9530, mAP50=0.9701, mAP50-95=0.6900
---------------------------------------------------------------------------
FileNotFoundError                         Traceback (most recent call last)
/tmp/ipykernel_981/188733028.py in <cell line: 0>()
    190 if __name__ == "__main__":
    191     pipe = PlateDetectorPipeline()
--> 192     pipe.run_all()

1 frames

/tmp/ipykernel_981/188733028.py in process_videos(self, sources, dst, conf, show)
    126         for i, cap in enumerate(caps):
    127             if not cap.isOpened():
--> 128                 raise FileNotFoundError(f"Не открывается видео: {sources[i]}")
    129
    130         fps = caps[0].get(cv2.CAP_PROP_FPS) or 25
```


CarLongPlateGen1622_jpg.rf
xemay816_jpg.rf.9297ac2438
CarLongPlateGen815_jpg.rf
CarLongPlateGen1428_jpg.rf
CarLongPlateGen3826_jpg.rf
CarLongPlateGen3106_jpg.rf
CarLongPlateGen3451_jpg.rf
CarLongPlateGen1114_jpg.rf
CarLongPlateGen752_jpg.rf
CarLongPlateGen1353_jpg.rf
ITS THE LAW




Найдено номеров: 1 (уверенность: 0.77)




**Вывод:** осуществил обучение нейросетевого детектора для решения задачи обнаружения номеров авто.